

Miles Raker

Aerospace Electrical Engineer · Active Secret Clearance

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PROFILE

Aerospace electrical engineer with experience taking programs from clean-sheet design through first flight at Scaled Composites. Company SME in flight test instrumentation, with hands-on depth in instrumentation, power distribution, payload networks, control systems, telemetry, PCB design, and custom Python test software.

TECHNICAL SKILLS

Instrumentation & Flight Test: IADS/RTS · DAS Studio · Curtiss-Wright ACRA KAM-500 · ACRA NET/SWI/101 · Parvus DuraNET · Quasonix · Silvus · Starlink · Starshield

Electrical Engineering: Power distribution design (AC 43.13) · Electrical load analysis · Control system design · PCB design · Altium · Harness design · Siemens NX · Mechanical design & fabrication

Networking & Protocols: GigE Vision · ARINC 429 · iNet-X · IENA · UDP · TCP · Multicast · Network design and validation

Software & Tools: Python · C# · C++ · C · MATLAB · Verilog · React · Java/Spring · Azure · AWS · Jira · Microsoft Project

PROFESSIONAL EXPERIENCE

Scaled Composites — Mojave, CA

Design Engineer | March 2023 – Present

Electrical design, flight test instrumentation, payload networks, and technical leadership across 20+ programs and 50+ flight tests.

FLIGHT TEST & INSTRUMENTATION

- For Vanguard's first-flight instrumentation package, designed the architecture, selected 150+ sensors, configured DAQ/network hardware, authored subsystem schematics and procedures, and mapped 2,500+ IADS parameters with real-time displays, giving test teams comprehensive system visibility through first flight.
- During **50+ flight tests**, served as dual-role electrical and telemetry control room operator, configuring ground station antenna, IADS server, and data routing while monitoring aircraft electrical parameters, troubleshooting anomalies in real time, and calling **knock-it-off** during Vanguard approach-to-stall testing after detecting beta limit exceedance.

ELECTRICAL DESIGN & PAYLOAD SYSTEMS

- For Vanguard's 18-month build-to-first-flight program, engineered relay-based **landing gear sequencing** and **fuel tank selection control systems** from clean sheet through design, fabrication, and validation.
- Across Vanguard and 6 payload programs, designed 28/12/5 VDC power distribution systems per AC 43.13 and maintained electrical load analysis documents for 5 aircraft; during customer payload integration, caught undersized wiring and circuit protection and required corrective upsizing before flight approval.
- Across 20+ payload integration programs, served as a company **subject matter expert for aircraft instrumentation** and configured payload network systems supporting classified and unclassified test operations.
- For Deimos and Phobos payload testbeds, redesigned the **fuel selection PCB and microprocessor** system, then built an acceptance test rig and authored formal procedures to validate system performance before integration.
- For a customer-specific payload network, developed bespoke **Python test software** to exercise network interfaces, support integration troubleshooting, and validate mission-specific test needs.

TECHNICAL LEADERSHIP & SAFETY

- For a multi-partner payload integration program, served as **technical lead** for the primary customer and 3 mission partners, coordinating development support, presenting at formal program reviews (PDR/CDR/TRR/FRR), authoring network design documents, ICDs, system descriptions, operation guides, and formal test procedures, and mentoring 2 new-hire electrical engineers.
- As a **signatory authority** for instrumentation and payload preflight readiness, conducted 100+ inspections, performed corrective rework on approximately 10% of inspected systems, and used checkout procedures to identify safety-critical discrepancies before 4 flights.

Federal Aviation Administration — Des Moines, WA

Electronics Engineer | September 2019 – January 2021

- For a waypoint navigation equipment upgrade, led engineering and field execution by completing environmental, airspace, and safety analyses, designing upgrade schematics, coordinating ATC service outages, and performing field modifications; delivered ahead of schedule and 10% under budget with zero navigation signal loss or injuries.
- To standardize DME project engineering reports, designed a survey template adopted across 63 projects, reducing engineering time by 10% while improving report clarity and accuracy.

US Army National Guard — Camp Murray, WA

Musician, Sergeant | February 2016 – January 2022

- During 87 days of COVID-19 response activation, led a 6-soldier squad operating a regional food bank under safety protocols and conducted contact tracing for 30+ cases.
- To improve unit technical readiness, delivered 3-part audio engineering training for 43 soldiers, contributing to a 50% reduction in wiring errors the following year and selection for promotion to Sergeant ahead of 8 peers.

US Army — Seoul, South Korea / Newport News, VA

Musician, Specialist | April 2009 – February 2016

- Led an international U.S./Korean team through design, rehearsal, and performance of 12 musical outreach missions for 15,000+ citizens in Seoul, South Korea.
- Built a stage lighting department from concept to sustainment, developing SOPs, securing resources, and establishing permanent staffing; expanded after-dark outreach capability and improved unit mission readiness.

EDUCATION

University of Washington	BS Electrical Engineering Cum Laude, 3.7 GPA Focus: microprocessors	2019
Washington State University	BM Music Performance Cum Laude, 3.6 GPA	2009
Codeup / St. Martin's University / MSSA	Full-Stack Java & Cloud Application Design	2021–2023

CERTIFICATIONS

- Electrical Engineer in Training (EIT)
- Curtiss-Wright IADS Training
- Amazon Web Services (AWS) Cloud Practitioner
- Microsoft Azure Fundamentals | Azure Data Fundamentals | Azure AI Fundamentals
- Linux Professional Institute LPI Le-1 Linux Essentials